Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **C++/Java systems**, **client-server applications**, and reliable software through OOP, SQL, testing, debugging, CI/CD, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit states, failure isolation, recoverable processing, and observable workflows under sustained production load reliably daily.
- Reduced dashboard latency from 1.2s to 350ms by optimizing **indexed SQL queries** and multi-tenant data models supporting 500+ users with normalized schemas and efficient access patterns reliably.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive customer workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **client-server platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time state synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, coupon validation, payment webhooks, and Firebase status synchronization while using transactions and locking to prevent conflicting reservations.
- Reduced manual refund handling by approximately 70% through **payment workflow automation**, supporting refunds, disputes, and consistent status while presenting architecture and backend integration workflows directly to clients reliably.

Skills

Languages: C++, Java, Python, SQL, JavaScript, TypeScript, Bash
Architecture & Systems: Object-Oriented Design, Client-Server Architecture, Data Structures, Algorithms, Operating Systems, Concurrency, Multithreading, Process Scheduling
Backend & Web: Node.js, React, Flask, FastAPI, REST APIs, WebSockets, Microservices, API Design
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Azure SQL, AWS RDS, AWS S3, AWS Glue, Docker
Engineering Quality: Git, GitHub, Jenkins, CI/CD, Unit Testing, Integration Testing, PyTest, Mockito, Debugging, Performance Optimization

Projects

Shared Game Services Library - Design Concept

- Designed a **C++ shared-library concept** with object-oriented interfaces for sessions, configuration, telemetry, persistence, and error handling, supported by dependency injection, unit tests, semantic versioning, and CI validation.

Client-Server Game State Monitor - Design Concept

- Designed a **client-server monitoring concept** using C++, SQL, React, and WebSockets to stream state transitions, expose diagnostic timelines, profile latency, isolate failures, and support replayable integration tests.

CPU Scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** for FCFS, SJF, SRT, Round Robin, priority, and MLQ, with edge-case handling, process-state visualization, and dynamic turnaround and waiting-time calculations across simulations reliably.